Patient Shortage in Histopathology: Breadth, Depth, and Effective Support

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Abstract

Training on fewer patients impairs histopathology patch classification even when class patch counts are held fixed. This experiment separates patient breadth from patch depth on a three-by-three allocation grid using BRACS and TCGA-UT frozen Virchow2 features and fixed patient-disjoint splits. Within each class and training split, all cells draw from one common pool of patients able to supply the maximum depth of 32 patches. At fixed breadth, repeated draws use identical patients across depths and nested patch subsets, so increasing depth reveals additional material from the same patients. Equal-budget comparisons and redundancy-adjusted support surfaces test whether additional patients contribute information beyond within-patient redundancy. Results under this common-pool design remain pending; estimates from depth-dependent eligibility do not identify these contrasts.

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1 Motivation

The signal taxonomy of Queisler (2026c) separates *nominal support*, the number of labelled patches n_c available for class c, from *independent support*, the number of distinct patients G_c supplying them. Patches from one patient share acquisition conditions and much of their morphology, so a large patch count assembled from few patients need not carry the evidence its size suggests. The same taxonomy already anticipates a quantitative version of this argument. Its *redundancy-adjusted effective support* $N_{\mathrm{eff},c} = n_c/\mathrm{DE}_c$ divides the patch count by a design effect $\mathrm{DE}_c = 1+(m_c^\star - 1)\,\mathrm{ICC}_c$, where $m_c^\star$ is the size-biased number of patches per patient and ICC_c is the within-patient correlation of a scalar summary of the representation (Kish, 1965; Rutterford et al., 2015).

Three earlier results define what is left to explain. The controlled shortage benchmark reduced patient support while holding class patch counts fixed and measured discrimination losses of 2.62–3.03 percentage points on BRACS and 4.76–4.83 points on TCGA-UT, which the tested mitigation methods recovered by between −6.5% and 2.5% (Queisler, 2026b). The classifier comparison then showed that the deficit is not a property of the downstream model: replacing the multilayer perceptron by multinomial logistic regression or cosine nearest neighbours improved concentrated-support accuracy by at most 0.48 points, while broader coverage benefited every classifier (Queisler, 2026e). The influence comparison finally showed that the deficit is not an artefact of unequal loss contributions: patient-average training changed accuracy by −1.36 points on BRACS and −0.004 points on TCGA-UT, leaving residual coverage benefits of 2.73 and 4.70 points (Queisler, 2026d). The bottleneck has thus been localized to the training material itself, but its content has not been characterized.

Two readings of that material remain. Under *within-patient redundancy*, patches from one patient are statistically dependent, so a patient's contribution is worth fewer independent observations than its patch count states; the deficit is then a shortage of effective observations and should be summarized by a single scalar. Under *coverage beyond redundancy*, patients differ in ways that no scalar correlation captures, because additional patients introduce morphological or acquisition modes absent from the observed ones; the deficit is then a shortage of distinct conditions rather than of effective observations. These readings are not mutually exclusive, and both are compatible with every result reported so far, because the previous allocations varied patient count and within-patient depth together.

Separating them requires an allocation in which the number of contributing patients and the number of patches taken from each of them move independently. This experiment therefore constructs a *breadth* × *depth* grid, with breadth G the number of patients contributing to a class and depth m the number of patches drawn from each of them, and compares allocations that differ in composition at an identical patch budget. Holding depth equal across patients has a second benefit: it removes the contribution inequality studied by Queisler (2026d) by construction, so patch-average and patient-average training coincide and the two objectives need not be compared again.

2 Study design

BRACS retains its seven lesion classes and TCGA-UT its 30 eligible cancer types. Both keep the benchmark dataset eligibility rules, class order, and frozen 2560-dimensional Virchow2 features of the benchmark, formed by concatenating the CLS token and the mean patch token (Zimmermann et al., 2024; Queisler, 2026a). The three existing patient-disjoint train–validation–test splits remain fixed, and BRACS patients and TCGA-UT participants are called patients throughout. Validation and test partitions keep their natural distributions and are never resampled.

Each training allocation is described by one pair (G, m) applied to every class: exactly G patients contribute to class c, and each contributes exactly m patches of that class, so $n_c = Gm$

for all classes. Class patch counts are therefore equal by construction, which removes prevalence as a competing signal and leaves support as the only quantity that varies. Table 1 lists the nine cells of the grid and their per-class patch budgets. Breadth ranges over $G \in \{5, 10, 20\}$ and depth over $m \in \{8, 16, 32\}$, a fourfold range in each factor and a sixteenfold range in n_c. The largest cell uses 4,480 training patches on BRACS and 19,200 on TCGA-UT.

	Patches per patient m		
Patients per class G	8	16	32
5	40	80	160
10	80	160	320
20	160	320	640

Table 1: Per-class patch budget $n_c = Gm$ for the nine allocation cells. Cells sharing a budget form equal-budget sets: $n_c = 160$ contains $(20, 8)$, $(10, 16)$, and $(5, 32)$, while $n_c = 80$ and $n_c = 320$ contain two cells each. Comparisons within such a set differ only in how a fixed number of patches is distributed over patients.

The proposed grid ceiling is set by the rarest class rather than by cost. The benchmark's broad-coverage allocations contain at least 30 patients per class, but this does not establish how many can supply 32 patches in each training split (Queisler, 2026d). Feasibility therefore requires a separate audit of the common maximum-depth pools. The largest proposed budget remains below the benchmark's balanced allocations of about 2,400 patches per class on BRACS and about 1,494 on TCGA-UT. Absolute accuracies are not directly comparable to those earlier reports; the object of interest is the shape of the support surface.

2.1 Allocation rules

Let N_{irc} denote the number of available patches of class c from patient i in the training partition of split r. Eligibility is fixed before any cell is sampled:

$$\mathcal{E}_{rc} = \{i : N_{irc} \geq m_{\max}\}, \qquad m_{\max} = \max\{8, 16, 32\} = 32. \tag{1}$$

Every breadth–depth cell uses this same class–split pool, including depths 8 and 16. Patients with fewer than 32 patches of the class are excluded from all cells. Otherwise, changing depth could change the distribution of tissue availability, lesion extent, slide count, or morphology among eligible patients, confounding a depth contrast with patient composition.

Before fitting, the preflight audit records $|\mathcal{E}_{rc}|$ for every class and split and requires $|\mathcal{E}_{rc}| \geq 20$ throughout. If this fails, execution stops and the entire three-by-three grid must be reduced consistently before a fresh preflight; no class, split, or cell receives a separate eligibility threshold or allocation ladder. Breadth is reduced first while retaining the common 32-patch threshold. If the maximum depth must also be reduced, eligibility is redefined once at that maximum for every cell. Realized ladders, pool counts, patch budgets, and contrast endpoints must be documented before fitting.

For each split, class, breadth G, and repeated draw d, G patients are sampled without replacement from $\mathcal{E}_{rc}$. The patient seed does not depend on the cell depth, so these exact patients contribute at all three depths within the draw. Classes are sampled independently; a patient eligible for several classes may contribute to more than one. For each selected patient, patches follow one deterministic round-robin ordering across slides, with stable slide and patch ordering. Each depth reveals a prefix of that same ordering, giving

$$\mathcal{P}_{ircd,8} \subset \mathcal{P}_{ircd,16} \subset \mathcal{P}_{ircd,32}. \tag{2}$$

Thus a fixed-breadth depth contrast changes only how many patches are revealed from the same patients under this selection rule. Each cell uses $R = 5$ repeated patient draws; every cell quantity averages over these draws, with corresponding draw indices paired across depths. Between-draw dispersion is reported separately as a descriptive measure of sensitivity to patient composition.

The readout is the deterministic multinomial logistic regression of Queisler (2026e), with an unpenalized intercept and the same feature preprocessing and convergence criteria. Its regularization strength is selected on the validation partition separately for every split, cell, and draw from the inherited grid, because the training-set size varies by a factor of sixteen across the grid and a single inherited value would confound support with regularization. Only converged fits enter the analysis. The multilayer perceptron and the nearest-neighbour readout are omitted: Queisler (2026e) established that the coverage effect is present under all three, and a single deterministic readout keeps the grid small enough to be run without repeated initializations.

3 Effective support

Equal depth simplifies the redundancy adjustment considerably. With every contributing patient supplying m patches, the size-biased patient size $m_c^\star = \sum_g m_{gc}^2/n_c$ equals m exactly, so the design effect and the effective support of Queisler (2026c) become

$$\mathrm{DE}_c(m) = 1+(m-1)\,\mathrm{ICC}_c, \qquad N_{\mathrm{eff},c}(G,m) = \frac{Gm}{1+(m-1)\,\mathrm{ICC}_c}, \qquad \lim_{m\to\infty} N_{\mathrm{eff},c} = \frac{G}{\mathrm{ICC}_c}. \quad (3)$$

The two grid factors therefore enter effective support asymmetrically. Breadth scales it proportionally, whereas depth scales it only until the design effect grows with m, after which additional patches from the same patients contribute progressively less. The limit makes the prediction concrete: under the redundancy account no depth can raise effective support beyond $G/\,\mathrm{ICC}_c$, and the transition occurs near $m \approx 1/\,\mathrm{ICC}_c$. Whether that transition is visible within the depth ladder is an empirical question about the representation, and one that the estimated correlations settle before any classifier is fitted.

3.1 Estimating the intraclass correlation

The correlation is estimated per class and split from the training partition only, following the procedure fixed in the taxonomy. Each frozen Virchow2 embedding is projected onto the leading principal direction of a principal component analysis fitted jointly across classes on a class-balanced reference sample, which yields a scalar score that does not depend on any fitted classifier. The one-way random-effects estimator with the unequal-cluster-size correction is then applied to these scores with patients as clusters (Kenny and La Voie, 1985), from a capped sample of at most 200 patients and 200 patches per patient. A cell-level value is obtained by averaging $N_{\mathrm{eff,c}}$ over classes.

Reporting these correlations serves a purpose beyond the support surface. Each ICC_c is the share of projected feature variance that lies between patients rather than within them, so the class-level table is a direct measurement of intra- and inter-patient heterogeneity for every lesion type and cancer type. Classes whose appearance varies strongly within a patient should have low correlations and lose comparatively little from concentrated sampling, and comparing these estimates with the class-level accuracy effects is an exploratory part of the analysis.

Equation (3) lets the three accounts of Section 1 be stated as separable predictions about the grid, summarized in Table 2. The predictions are distinguishable because $n = Gm$, G, and N_{eff} are different functions of the two grid factors: an equal-budget set holds n constant while G and N_{eff} still vary across its cells.

Account	Claim about the deficit	Prediction on the grid
Nominal support	Only the amount of labelled material matters	Accuracy is a function of $n = Gm$; equal-budget cells agree
Redundancy-adjusted support	Within-patient patches are correlated, so they are worth fewer independent observations	Accuracy is a function of N_{eff}; depth saturates near $m \approx 1/\,\text{ICC}$; cells collapse onto one curve
Coverage beyond redundancy	Patients supply variation that a scalar correlation cannot express	Breadth remains beneficial at matched N_{eff}; a positive residual breadth term persists

Table 2: The three accounts and the grid behaviour each of them implies. The accounts are ordered by how much structure they attribute to the patient grouping. The first is already disfavoured by the benchmark result and is retained as a reference line within this design.

4 Evaluation

The primary endpoint is *patient-macro balanced accuracy*, defined exactly as in the two preceding experiments so that effect sizes remain comparable. For split r, let $\mathcal{P}_{rc}$ contain the test patients with at least one patch of true class c, and let $\mathcal{J}^{\text{test}}_{irc}$ contain that patient's evaluated patches of that class. The split score and the underlying within-patient recalls are

$$A_r = \frac{1}{K}\sum_{c=1}^{K}\frac{1}{|\mathcal{P}_{rc}|}\sum_{i\in\mathcal{P}_{rc}} q_{irc}, \qquad q_{irc} = \frac{1}{|\mathcal{J}^{\text{test}}_{irc}|}\sum_{j\in\mathcal{J}^{\text{test}}_{irc}} \mathbf{1}\{\widehat{y}_j = c\}, \tag{4}$$

where K is the number of classes. A cell score $A(G, m)$ averages the three split scores and the R draws equally. Secondary endpoints are macro negative log-likelihood in nats and expected calibration error over ten equal-width confidence bins, which connect this grid to the probability-quality damage established by Queisler (2026b), together with patch-micro balanced accuracy and class-specific patient-macro recalls. Probabilities are evaluated as produced; post-hoc temperature scaling is not applied.

Three contrasts summarize the surface, all expressed in percentage points:

$$\Delta_m(G) = A(G, 32) - A(G, 8), \qquad \text{depth gain at fixed breadth,} \tag{5}$$

$$\Delta_G(m) = A(20, m) - A(5, m), \qquad \text{breadth gain at fixed depth,} \tag{6}$$

$$X = A(20, 8) - A(5, 32), \qquad \text{equal-budget breadth advantage.} \tag{7}$$

The equal-budget advantage X compares the broadest and the deepest allocation of the $n_c = 160$ set and is the most informative single quantity in the design, because both of its terms consume the same labelling budget and differ only in how that budget is spread over patients. The intermediate cell $(10, 16)$ completes the set and indicates whether the exchange between breadth and depth is monotone.

4.1 Support surface

The contrasts describe individual comparisons, whereas the accounts of Table 2 concern the grid as a whole. Cell accuracies are therefore also summarized by least-squares fits over the nine cells, using the candidate predictors of Equation (3) on a logarithmic scale,

$$A = \alpha + \beta \log n, \qquad A = \alpha + \beta \log G, \qquad A = \alpha + \beta \log N_{\text{eff}}, \tag{8}$$

which have equal numbers of parameters and can be ranked directly by residual standard deviation. Two augmented fits then separate the accounts. Adding breadth to the nominal predictor, $A = \alpha + \beta \log n + \gamma_n \log G$, asks whether patient count matters at all once the patch budget is known, and so reproduces the benchmark finding within this design. Adding breadth to the effective predictor, $A = \alpha + \beta \log N_{\text{eff}} + \gamma_e \log G$, asks the question this experiment exists for: whether patients still matter once their redundancy has been accounted for. The second fit nests the third model of Equation (8) at $\gamma_e = 0$.

Uncertainty follows the procedure of the two preceding experiments. Each of 2,000 replicates of a paired Bayesian bootstrap assigns random positive weights drawn from a Dirichlet$(1, \ldots, 1)$ distribution to the distinct test patients (Rubin, 1981), with every patient's weight shared across cells, draws, splits, classes, and repeated test appearances. Within-class recalls, cell scores, contrasts, and surface coefficients are recomputed on each replicate, and the 2.5th and 97.5th percentiles form individual exploratory 95% intervals without adjustment for multiple comparisons. These intervals describe held-out patient variation conditional on the realized training draws; variation from the draws themselves is reported separately as the between-draw dispersion of Section 2.1.

4.2 Interpretation criteria

One percentage point remains the prespecified practical-relevance threshold (Queisler, 2026e). A point estimate above one point is practically promising, a lower interval bound above one point supports an effect exceeding the threshold, and an interval crossing zero leaves the direction open. For the surface coefficients the threshold is expressed on the same scale by noting that doubling breadth changes $\log G$ by $\log 2$, so a coefficient matters practically when $\gamma \log 2$ exceeds one point, that is when γ exceeds 1.44 points per log unit.

The accounts are then read as follows. Support that depends only on the patch budget requires both X and γ_n to be indistinguishable from zero, which the benchmark result makes unlikely and which therefore functions as a check on the design rather than as a live hypothesis. Redundancy-adjusted support is supported when the effective predictor attains the smallest residual standard deviation of the three fits in Equation (8) and the interval for γ_e lies inside the practical band around zero. Coverage beyond redundancy is supported when the interval for γ_e excludes zero with a positive point estimate, and is practically relevant when that estimate also exceeds 1.44 points per log unit. Because both augmented fits are conditional on the estimated correlations, a positive γ_e can also indicate that the scalar projection underestimates within-patient dependence, so the class-level correlation table is consulted before the residual term is read as missing patient variation.

Depth saturation is judged separately from the fits. It is supported when $\Delta_m(20)$ has an interval contained in ± 1 point, that is when quadrupling the patches taken from each of twenty patients leaves accuracy practically unchanged. This criterion is informative only if the predicted transition falls inside the ladder, so the mean estimated correlation is compared with the depth levels in advance: a mean ICC below $1/32$ places the transition beyond the deepest cell, and an absence of saturation is then uninformative about the redundancy account rather than evidence against it.

5 Results status

No results from the common maximum-depth eligibility design are available yet. Existing estimates were generated with depth-dependent eligibility and depth-dependent patient draws. They cannot establish the depth gains, equal-budget breadth advantage, or residual breadth effects defined here, because patient composition could change with depth. All nine cells, their repeated draws, and the downstream contrasts and support surfaces require recomputation before numerical conclusions can be reported for this design.

6 Limitations

The common eligibility rule restricts training to patients with at least $m_{\max}$ patches of the relevant class. These patients may differ systematically from patients with less tissue or fewer slides. Pairing patients and nesting patches removes changes in eligibility and sampled patient identity from within-draw depth contrasts, but does not establish generalization to excluded patients. Validation and test populations remain unchanged. Depth effects also depend on the deterministic slide-balanced patch ordering used here.

The grid is bounded by the rarest class in each cohort, so its largest allocation stays well below the benchmark's balanced allocations and the absolute accuracies of this experiment are not comparable with those of the earlier reports. The design also fixes equal depth for every patient, which removes contribution inequality but makes the allocations less similar to naturally collected cohorts, where patients differ widely in how much material they supply. Conclusions about the shape of the support surface therefore transfer to naturally unequal cohorts only through the design-effect model.

The redundancy adjustment inherits the approximations of the taxonomy. Its correlation is estimated from a one-dimensional projection of the representation under an exchangeable within-patient correlation model, so dependence that the leading principal direction does not express is absorbed into the residual breadth term. The support surface itself is a descriptive least-squares summary over nine cells; it establishes which predictor orders the observations, not a causal or generative account of the learning problem. All estimates are conditional on frozen Virchow2 features, on multinomial logistic regression, and on the three fixed splits, whose partitions overlap across splits.

Finally, the experiment locates the deficit in the training material but cannot say whether the missing information is absent from the cohort or merely inexpressible in this representation. A residual breadth term is compatible with both, and separating them requires varying the representation rather than the allocation. Probability quality is evaluated without temperature scaling, so the secondary calibration endpoints describe the fitted probabilities rather than the best attainable ones.

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